3/8/16 Discussion notes

Drude model:

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = \frac{1-\omega_0^2}{\omega(\omega+i\gamma)}$$

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 + \frac{i}{\omega}\sigma(\omega)$$

$$\omega_0^2 = \frac{e^2 n}{\varepsilon_0 m} \Big\} \text{ plasma frequency}$$

	$\varepsilon''(\sigma')$
good, dirty metals (reflection)	opaque dielectric media (absorption)
good, clean metals	transparent dielectric media

$\varepsilon'(\sigma'')$

dirty $\rightarrow$ large γ (scattering angle)
clean $\rightarrow$ small γ

good $\rightarrow$ large n
bad $\rightarrow$ small n

Surface waves

4 Fields: $\vec{E}$, $\vec{D} = \varepsilon(\omega)\vec{E}$

$\vec{B}$, $\vec{H} = \vec{B}/\mu_0$ } assuming non-magnetic material

$$\nabla\cdot\vec{D} = \rho_f \qquad \nabla\times\vec{E} = \frac{d\vec{B}}{dt}$$

$$\nabla\times\vec{H} = j_f + \frac{d\vec{D}}{dt}$$

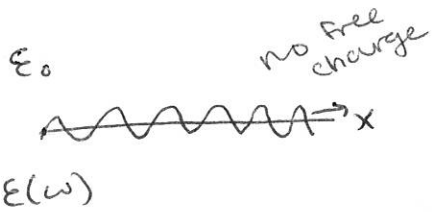
$\downarrow$

$\vec{D}$ will be continuous unless ρ_f is a delta function

$$n = \sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}}$$

$\longrightarrow$ look at a wave propagating on the boundary

$$\vec{E}, \vec{B} \sim e^{ikx - i\omega t} \Big\} \omega = \omega(k)$$



} amplitude will decay away from surface.

$$\left.\begin{aligned}\nabla\times\vec{E} &= i\omega\vec{B}\\ \nabla\times\vec{B} &= \mu_0\vec{j} - \frac{i\omega}{c^2}\vec{E}\end{aligned}\right\} \text{vacuum}$$

$$\text{bound currents/charges} \left\{\begin{aligned}\nabla\times\vec{E} &= -i\omega\mu_0\varepsilon(\omega)\vec{E}\\ \nabla\times\vec{B} &= -\frac{i\omega}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\vec{E}\end{aligned}\right.$$

$$\vec{E} = (E_x, 0, E_z)$$

$$\vec{B} = (0, B_y, 0)$$

① Get equation for $\vec{E}$ only

② solve it for a solution decaying away from surface

③ Find $\vec{B}$ from ②

④ satisfy boundary conditions

① $\nabla \times \vec{E} = i\omega \vec{B}$

$$\underbrace{\nabla \times (\nabla \times \vec{E})} = i\omega(\nabla \times \vec{B}) = i\omega\left(-\frac{i\omega}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\vec{E}\right)$$

$$\nabla(\underbrace{\cancel{\nabla \cdot \vec{E}}}_{0}) - \nabla^2 \vec{E} = \frac{\omega^2 \varepsilon(\omega)}{c^2 \varepsilon_0}\vec{E}$$

(no bound charges)

$$\nabla^2 \vec{E} = -\frac{\omega^2 \varepsilon(\omega)}{c^2 \varepsilon_0}\vec{E} = \frac{d^2 E(z)}{dz^2} - k^2 E(z)$$

$$\vec{E} = \vec{E}(z) e^{ikx - i\omega t}$$

negative so that it will decay away from the boundary

$$E_x(z>0) = e^{-z\sqrt{k^2 - \omega^2/c^2}}$$

$$E_x(z<0) = e^{z\left(k^2 - \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\right)}$$

at this point we have satisfied one equation for E_x, one boundary condition

$$-\frac{i\omega}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}E_x = \cancel{\frac{dB_z}{dy}}^{\,0} - \frac{dB_y}{dz} \longrightarrow \frac{dB_y}{dz} = \frac{i\omega}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}E_x$$

$B_y(z)$ will have same z-dependence as

$$E_x(z) \longrightarrow B_y(z>0) \sim e^{-z\sqrt{k^2 - \omega^2/c^2}}$$

$$B_y(z<0) \sim e^{z\left(k^2 - \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\right)}$$

$$\frac{dB_y(z>0)}{dz} = \frac{i\omega}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0} e^{-z\sqrt{k^2-\omega^2/c^2}}$$

$$B_y(z>0) = \frac{i\omega\,\varepsilon(\omega)}{c^2\varepsilon_0}\int e^{-z\sqrt{k^2-\omega^2/c^2}}\,dz = \frac{-i\omega\,\varepsilon(\omega)}{c^2\varepsilon_0\sqrt{k^2-\frac{\omega^2}{c^2}}}\, e^{-z\sqrt{k^2-\omega^2/c^2}}$$

$$\frac{dB_y(z<0)}{dz} = \frac{i\omega}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0} e^{z\left(k^2-\frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\right)}$$

$$B_y(z<0) = \frac{i\omega\,\varepsilon(\omega)}{c^2\varepsilon_0\left(k^2-\frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\right)}\, e^{z\left(k^2-\frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\right)}$$

$$\frac{\varepsilon(\omega)}{\varepsilon_0}\sqrt{k^2-\frac{\omega^2}{c^2}} = -\sqrt{k^2-\frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}} \longrightarrow \therefore\ \varepsilon(\omega) \text{ must be real, negative}$$

- will continue with this problem in 3/10 discussion